Results & Discussion draft — What Drives an LLM's Choice? (Chris Crasto Gomes)

5. Results

All figures below are computed from the completed trial records (`results/results.csv` , 1,041 rows; `results/results_run3.csv` , 206 rows) under the denominator policy of §4.5: percentages use completed trials, dropped trials are reported as counts out of the planned total, and every percentage is accompanied by its count.

5.1 Descriptives — did the redesign work?

Run 1 produced a ceiling effect (Artistry Academy won 96.6% of trials) because its winning page stacked every top driver at once; runs 2a and 2b then over-corrected in the opposite direction (a review-count knockout, then a Canvas page that won nearly everything). Both were aborted at the design stage (§4.6). The question for the completed Run 2 is therefore whether the rebalanced design finally produced the intended five-way spread without re-introducing dominance.

It removed dominance but did not produce five-way balance. Across all 1,041 Run-2 trials:

Page	Wins	Share
Artistry Academy	584	56.1%
Palette Masters	416	40.0%
Brushstroke Studio	24	2.3%
Creative Canvas	14	1.3%
Canvas & Color	3	0.3%

No page reaches the pre-registered 60% dominance threshold — the run-1 ceiling (96.6%) is resolved — but the distribution is two-strong rather than flat: Artistry and Palette together take 96.1% of all picks, while Canvas and Creative together take 1.6%.

Of the 1,041 trials, **444 ran on the v1 pages (16 Oct) and 597 on the final frozen v2 pages (17 Oct)** (§4.2). Canvas did not win a single v1 trial (0/444), and because v1 → v2 changed only the visual layer — the controlled values being byte-identical across the boundary — the v1 and v2 trials differ in markup, not in the data values the models reasoned about. All like-for-like comparisons against Run 3 use the **597 final-page trials**, which are additionally convenient because they consist exclusively of the five models that also ran in Run 3 (§5.5).

5.2 Per-model choice distributions

Model	N	Choice distribution	Signature
gemini-3.8-flash	148	A 93.9% (139) · P 6.1% (9)	credentials-first
mimo-v2.6-flash	150	A 84.0% (126) · P 16.0% (24)	credentials-first
deepseek-flash	143	A 82.5% (118) · P 17.5% (25)	credentials-first
grok-3-mini	150	P 72.0% (108) · A 18.7% (28) · Cr 8.7% (13) · B 0.7% (1)	price-first
gemini-3.1-flash-lite	150	P 69.3% (104) · A 18.0% (27) · B 12.7% (19)	price-first
gpt-6-luna	150	P 56.7% (85) · A 43.3% (65)	price-leaning, genuinely split
gpt-4.1-mini	150	A 54.0% (81) · P 40.7% (61) · B 2.7% (4) · C 2.0% (3) · Cr 0.7% (1)	heterogeneous (all five pages)

Three observations:

1. **The aggregate balance is between-model, not within-model.** At temperature 0.1 no model is anywhere near uniform over the five offers; each has a dominant pick between 54% and 94%. The pooled near-five-way spread at the top of §5.1 is produced by models disagreeing with *each other* — credentials-first models and price-first models cancelling out — not by any individual model weighing options evenly.

2. **Three heuristic signatures recur:** credentials-first (gemini-3.8-flash, mimo, deepseek — all anchor on Artistry's Yale MFA/MoMA profile), price-first (grok, gemini-lite, luna — anchor on Palette's $149), and one heterogeneous model (gpt-4.1-mini — the only model to pick all five pages, and the only model whose Run-2 Canvas picks are non-zero).
3. **The two weak pages fail for different reasons and are rescued by different models.** Brushstroke's weakness (fewest reviews, second-highest price) is only ever overlooked by gemini-lite (19 picks); Creative's gentle-pitch/installment bundle is only ever taken by grok (13 picks). Canvas — most reviews but lowest rating, no credentials, no certificate — is picked by exactly one model, once every ~50 trials (gpt-4.1-mini, 3/150).

5.3 Version sensitivity — the DeepSeek before/after

Three models ran on both page versions inside Run 2, which makes them a natural check on whether the visual-layer change (§4.2) moved choices. Their picks, before (v1, 16 Oct) and after (v2, 17 Oct):

Model	v1 picks	v2 picks	Shift
deepseek-flash	n=71: A 67 (94%) · P 4 (6%)	n=72: A 51 (71%) · P 21 (29%)	**–23pp Artistry**
gpt-6-luna	n=32: P 18 (56%) · A 14 (44%)	n=118: P 67 (57%) · A 51 (43%)	stable (±1pp)
gemini-3.8-flash	n=41: A 39 (95%) · P 2 (5%)	n=107: A 100 (93%) · P 7 (7%)	stable (2pp)

Two of the three dual-version models hold their signature across the change. **DeepSeek does not:** its Artistry share falls 23 percentage points and its Palette share roughly quintuples, even though every controlled value it could reason about is byte-identical between the two blocks. The most economical reading is that DeepSeek's choice is sensitive to the *presentation* layer (layout, imagery, markup structure) that the v2 redesign altered — a stimulus-version effect on a model that was otherwise one of the most stable in the pool.

The consequences are carried into every later claim: (i) part of the between-model variance observed in §5.2 is stimulus version, not model identity, so "all variance is between models" is stated only with this qualification; (ii) single-version models (gemini-lite, mimo — v1 only; gpt-4.1-mini, grok — v2 only) are version-pooled in §5.2 with that caveat attached; (iii) all between-run comparisons in §5.5 are restricted to the frozen v2/v3 pages; and (iv) from 17 Oct onward the pages are frozen, with Run 3 as the only sanctioned change.

5.4 Stated vs revealed choice

The exit survey (§4.3) is administered **after** the choice, in the same conversation, so its answers are post-choice self-report: they may simply rationalise a decision the model has already made. It is therefore analysed for **consistency between stated and revealed preference**, never as evidence that a rated factor caused the choice.

Consistency is high. Choosing models' survey ratings track their choices closely:

Choosers of…	N	Rate highest	Rate lowest
Artistry Academy	584	Credentials 4.99 · Reviews 4.94	Price 3.30
Palette Masters	416	Price 5.00	Credentials 2.51
Brushstroke Studio	24	Credentials 5.00 · Comparison 4.88	Reviews 3.46
Creative Canvas	14	Clarity 5.00 · Price 5.00	— (Payment plan 4.21)

Across all 1,041 trials, the stated-importance ordering is **Description 4.61 ≈ Reviews 4.56 > Clarity 4.41 > Price 4.03 ≈ Comparison 3.99 ≈ Credentials 3.96 > Social proof 3.10 > Badge 2.61 > Payment plan 2.24 > Anchoring 2.00.**

Two readings follow. First, what models *say* and what models *do* line up: the credentials- first models rate credentials highest, the price-first group rates price highest, and the never-chosen levers (badges, strike-through anchoring, installments) are also the levers models rate as least influential. Second, the classic conversion-optimisation toolkit — positioning badges, price anchoring, installment plans — barely registers as a stated driver for any model in this tier, while description detail and review counts dominate. Because the survey is post-choice, both readings are reported as association/consistency; the design cannot establish that the rated factor *caused* the pick.

5.5 Run 3 — rating as the single-variable test

Design. Run 3 changes exactly one feature: Canvas & Color's star rating 4.3★ → 4.6★ (raising it off the floor of the five; it now ties Palette at 4.6★). Every other feature, and the other four pages, are byte-identical to the frozen v2 pages (`pages_run3/` , diff-verified; §4.2). Protocol identical: temperature 0.1, all 7 models × 30 = 210 planned, **206 completed** (4 dropped, §4.5), rows tagged `page_version=v3` .

- **H1:** if rating was the binding constraint in Run 2, Canvas's share rises materially above its baseline.
- **H2:** it stays low → the bundle reading of §5.1–5.2 stands.

Headline — model set held constant. The cleanest comparison restricts to the five models that ran in both runs on the same (v2/v3) pages — gpt-6-luna, deepseek-flash, grok-3-mini, gpt-4.1-mini, gemini-3.8-flash. Their Run-2 final-page trials are exactly the 597, and their Run-3 trials are 146 of the 206:

	Run 2 (v2 pages)	Run 3 (v3 pages)	Shift
Canvas share, five models in both runs	**3 / 597 (0.5%)**	**45 / 146 (30.8%)**	**+30.3pp**

The two remaining models — mimo-v2.6-flash and gemini-3.1-flash-lite — ran Run 2 on the v1 pages and therefore have no same-pages baseline; their Run-3 results are reported separately: **Canvas 10 of 60 (16.7%)** (mimo 10/30, gemini-lite 0/30). Their comparison crosses the visual redesign *and* the rating change; because v1 → v2 altered only the visual layer, the data-value confound for them is nil, but the presentation-layer confound (which §5.3 showed can move a model) remains — so they are kept out of the headline and declared here.

Pooled across all seven models, Run 3 gives Canvas **55 of 206 (26.7%)**; the pooled figure is reported wherever the whole run is described, with the two groups identifiable.

Aggregate movement (all 206 Run-3 trials):

Page	Run 2 · final pages (n=597)	Run 3 (n=206)	Shift
Artistry Academy	311 (52.1%)	90 (43.7%)	–8.4pp
Palette Masters	264 (44.2%)	55 (26.7%)	–17.5pp
Canvas & Color	**3 (0.5%)**	**55 (26.7%)**	**+26.2pp**
Brushstroke Studio	5 (0.8%)	4 (1.9%)	+1.1pp
Creative Canvas	14 (2.3%)	2 (1.0%)	–1.3pp

Per model — Run 2 Canvas (pages seen) vs Run 3 counts:

Model	Run 2 · Canvas	Run 3 · counts	Reading
gpt-6-luna	0/118 (v2)	Canvas 17/30 (57%)	flipped outright
deepseek-flash	0/72 (v2)	Artistry 14/26 (54%) · Canvas 11/26 (42%)	split
grok-3-mini	0/150 (v2)	Palette 12/30 (40%) · Canvas 11/30 (37%)	split
mimo-v2.6-flash *	0/150 (v1)	Artistry 17/30 (57%) · Canvas 10/30 (33%)	moderate
gpt-4.1-mini	3/150 (v2)	Artistry 15/30 (50%) · Canvas 6/30 (20%)	moderate
gemini-3.1-flash-lite *	0/150 (v1)	Palette 24/30 (80%)	unmoved
gemini-3.8-flash	0/107 (v2)	Artistry 30/30 (100%)	unmoved

* Ran Run 2 on the v1 pages — no same-pages baseline; confound discussed above and in §7.

Reading — support for H1. The movement is structured exactly as the Run-2 signatures predict. Every price-leaning or heterogeneous model shifted toward Canvas the moment its rating ceased to be the lowest of the five: gpt-6-luna flipped outright (0/118 → 17/30), deepseek and grok split their votes between their original anchor and Canvas, and gpt-4.1-mini — already the only model picking Canvas in Run 2 — doubled its share. The two extreme-signature models did not move at all: gemini-3.8-flash chose Artistry 30/30, and gemini-lite stayed with price-first Palette (24/30). For the five like-for-like models this is **Canvas 3/597 (0.5%) → 45/146 (30.8%)** from a single feature change — which supports H1: rating was the binding constraint for the middle of the field.

The same result answers the bundle question of §5.1–5.2 (feedback point 3): review volume alone never won a trial for any model (Canvas's 260 reviews sat with the lowest rating and never won), but volume *plus* a non-lowest rating wins 30.8% for the like-for-like five — second place in the aggregate, not first. Volume was never powerless; it was masked by the rating floor. Canvas's remaining deficits (no credentials, no certificate) still cost it the credentials-first segment, which is why Artistry still leads Run 3 overall (43.7%) and the field reads **Artistry first, Palette and Canvas tied for second (26.7% each pooled)** — not a takeover, and still not five equal pages.

Evidential strength. With ≈30 trials per model (26 for deepseek), these are tendencies, not precise effect estimates; counts are given beside every percentage and the claim is worded throughout as *support for H1*, never confirmation or causal isolation.

Cost and failures. Run 3 cost **$6.82** for the 206 completed trials; the 4 dropped trials were all DeepSeek empty-response flakes (the same failure mode as Run 2, at a higher per-trial rate — a second resume pass recovered 6 of the first 10), counted out of 210 planned and never imputed (§4.5).

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6. Discussion

6.1 Main finding, properly hedged

Seven low-tier models, shown the same five controlled offers, do not behave like a single “LLM buyer”. They behave like a small population of consistent heuristic shoppers: within a model, choices are near-deterministic at temperature 0.1; between models, choices disagree sharply — credentials-first (three models), price-first (three models), one heterogeneous generalist. The aggregate therefore shows no dominance (no page ≥60%), but that balance is a *census of disagreeing models*, not evidence of per-model uncertainty. Two qualifications are declared rather than smoothed away: stimulus version can move a model (DeepSeek’s 23pp shift, §5.3), so part of the between-model spread reflects which page version each model saw; and ≈30 trials per model in Run 3 means all run-3 statements are tendencies at the level of counts, not point estimates.

6.2 What models heeded — and what they ignored

The stated-driver ordering (§5.4) and the choices point the same way. Models heeded **description detail (4.61)**, **review counts (4.56)**, clarity of structure (4.41), and — split by signature — either price (4.03) or credentials (3.96). The classic conversion- optimisation levers barely registered: badge 2.61, installment plan 2.24, price anchoring 2.00 — and the choices confirm the irrelevance, since no page won by stacking them. For this model tier, a positioning badge or a strike-through “was $349” is close to invisible; a concrete, legible price and a coherent credential or review story is not. The consistency between the two (stated ≈ revealed) is itself a result: unlike human survey respondents, these models’ post-hoc explanations are not obviously confabulated — though the post-choice design means the agreement is reported as consistency, never as causal evidence.

6.3 Review volume: a bundle, not an isolated driver

Run 2’s starkest number — Canvas 3 of 1,041 (0.3%), despite carrying the most reviews (260) — is a statement about **a bundle**: most reviews arriving together with the *lowest* rating of the five, no credentials, no certificate, community teaching. Run 3 was designed to cut that bundle apart by moving exactly one strand: rating 4.3★ → 4.6★, everything else byte-identical. For the five models present in both runs the effect is 0.5% → 30.8% (3/597 → 45/146). The reading that survives both runs is therefore: **review volume is a strong signal that was masked by a rating floor, not a weak signal**; volume plus a non-lowest rating is worth second place; and rating was the binding constraint for the middle of the field, while credentials still anchor first place for the credentials-first segment. H1 is supported, not confirmed — one rating change on one page, ~30 trials per model — and the two models without a same-pages baseline contribute 10 of 60 Canvas picks that are kept separate from the headline. What Run 3 cannot do is separate the *within-bundle* roles of credentials and certificate for Canvas (they were not manipulated), which is why claims about why Canvas still finishes behind Artistry remain associational.